ApproxDA-TransUNet: Understanding Global Context Sensitivity of Attention Approximation for Medical Image Segmentation

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***Abstract*—Dual-attention mechanisms have demonstrated strong performance in medical image segmentation by jointly modeling long-range spatial dependencies and channel relationships. However, their high computational cost has motivated numerous attention approximation methods, which primarily focus on improving efficiency while providing limited understanding of when approximation is beneficial across different segmentation tasks. To address this challenge, we propose ApproxDA-TransUNet, an efficient dual-attention approximation framework that replaces expensive global attention with controllable windowed low-rank spatial attention and grouped channel attention. The proposed framework provides explicit control over the approximation scale through window size, approximation rank, and channel grouping, enabling efficient segmentation while facilitating systematic analysis of approximation behavior.**

Extensive experiments on three representative medical image segmentation benchmarks, including Synapse, Kvasir-SEG, and ISIC 2018, demonstrate that ApproxDA-TransUNet consistently outperforms the original DA-TransUNet, achieving 80.94%, 90.17%, and 89.58% Dice, respectively. Beyond the performance improvements, our analysis reveals two key findings. First, the effectiveness of attention approximation is strongly task dependent. We observe that segmentation tasks exhibit different degrees of Global Context Sensitivity (GCS), which can be empirically characterized by their sensitivity to attention window size, explaining why different tasks favor different approximation scales. Second, we show that the commonly adopted learnable dual-attention gate consistently collapses toward nearly uniform routing because of gradient symmetry, limiting the benefit of adaptive feature fusion.

These findings suggest that attention approximation should be regarded not merely as an efficiency technique, but as a task-dependent inductive bias whose effectiveness is governed by the contextual requirements of the target segmentation problem. Our work provides both an effective segmentation framework and new insights into understanding attention approximation in medical image segmentation.

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, global context sensitivity, gate collapse**

I. Introduction

Transformers have revolutionized medical image segmentation by enabling global long-range dependency modeling that convolutional networks cannot achieve. Dual attention [1]—combining a Position Attention Module (PAM) for spatial correlations and a Channel Attention Module (CAM) for channel correlations—emerged as a powerful mechanism, and methods such as DA-TransUNet [2] demonstrated its effectiveness in hybrid CNN-Transformer architectures for multi-organ and polyp segmentation. The quadratic complexity of dual attention, however, has prompted a wave of efficient approximations—windowed attention [3], low-rank factorization [4], and grouped channel interaction [5]—that are now standard design choices. These strategies share an implicit assumption that approximation is universally applicable with minor accuracy trade-offs—an assumption rarely examined across tasks with fundamentally different spatial reasoning requirements:

When, and at what scale, is attention approximation beneficial for medical image segmentation?

To investigate this, we design **ApproxDA-TransUNet**[1]: a dual-attention architecture with three orthogonal, independently controllable approximation operators—windowed PAM, low-rank projection, and grouped CAM. ApproxDA-TransUNet serves as both a practical efficient model that improves over DA-TransUNet on all three benchmarks, and an interpretable experimental platform for studying when and why approximation helps.

Applying ApproxDA-TransUNet across three medical segmentation benchmarks reveals a structured pattern: the same windowed attention operator is highly window-sensitive on multi-organ CT yet largely window-robust on polyp and skin lesion segmentation. Systematic ablation confirms that appropriate window selection consistently benefits all three benchmarks, but the degree of sensitivity differs markedly—tasks with high spatial context requirements prove $3.6\times$ more window-sensitive than those dominated by local appearance. We further find that learnable routing collapses to a fixed equilibrium due to gradient symmetry—a structural property of symmetric two-branch architectures, not a training artifact.

The main contributions of this paper are:

- **ApproxDA-TransUNet**: We propose ApproxDA-TransUNet, an efficient dual-attention approximation

[1]Source code will be made available upon acceptance.

architecture with three orthogonal controllable axes—windowed PAM, low-rank projection, and grouped CAM—that consistently improves segmentation performance over the original DA-TransUNet and competitive Transformer-based baselines.
- **Empirical discovery of GCS**: Using ApproxDA-TransUNet as a controllable experimental platform, we present a systematic empirical study of attention approximation across heterogeneous medical segmentation tasks, uncovering a 3.6× difference in window sensitivity between Synapse and Kvasir-SEG. We term this task-level variability *Global Context Sensitivity (GCS)* and demonstrate that it predicts window-tuning importance across tasks.
- **Mechanistic insight**: We provide empirical evidence suggesting that attention approximation introduces a task-dependent inductive bias better aligned with tasks of lower contextual requirements, and further observe gradient symmetry collapse: learnable dual-branch routing degenerates to $g{\approx}0.5$ regardless of window size—a structural limitation of symmetric two-branch gating.

The pattern we uncover is structured by GCS: tasks with high contextual requirements demand careful window tuning; those with low requirements are largely window-robust. The underlying mechanism is an inductive bias shift—approximation does not just reduce cost, it changes what the attention mechanism is sensitive to.
The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA-TransUNet framework; Section IV reports results against state-of-the-art methods; Section V provides mechanistic analysis; Section VI concludes.

II. Related Work

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] and its successors [7]–[10] progressively refining skip connections, attention gating, and multi-scale feature representations. Despite strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], and DAEformer [15] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks. These methods treat global context modeling as uniformly beneficial, however, without examining whether different tasks require different degrees of spatial context.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, achieving state-of-the-art accuracy while inheriting the quadratic cost of full-map attention.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [5], [15]. Existing efficient attention methods primarily optimize computational efficiency while implicitly assuming approximation is universally applicable. Whether approximation itself changes representation learning, and why it behaves differently across tasks with different contextual requirements, remains largely unexplored. This work addresses both questions by studying the context-sensitive behavior of approximation across heterogeneous segmentation tasks and exposing the gradient symmetry dynamics that cause adaptive routing to degenerate into fixed blending.

III. Method

This section introduces the overall architecture of ApproxDA-TransUNet, then describes the three controllable approximation operators used to reformulate the dual-attention module, and finally analyzes their computational complexity.

A. Architecture Overview

Figure 1 illustrates the overall architecture: ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes $14{\times}14$ patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28, 256 at 56×56, and 64 at 112×112—and at the bottleneck. Each block routes activations through a Low-Rank Windowed PAM branch and a Grouped CAM branch, fused by a configurable gate g that controls the spatial-to-channel attention blend.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA-TransUNet reduces this cost along three independent axes (Figure 2), each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M{\times}M$ local windows

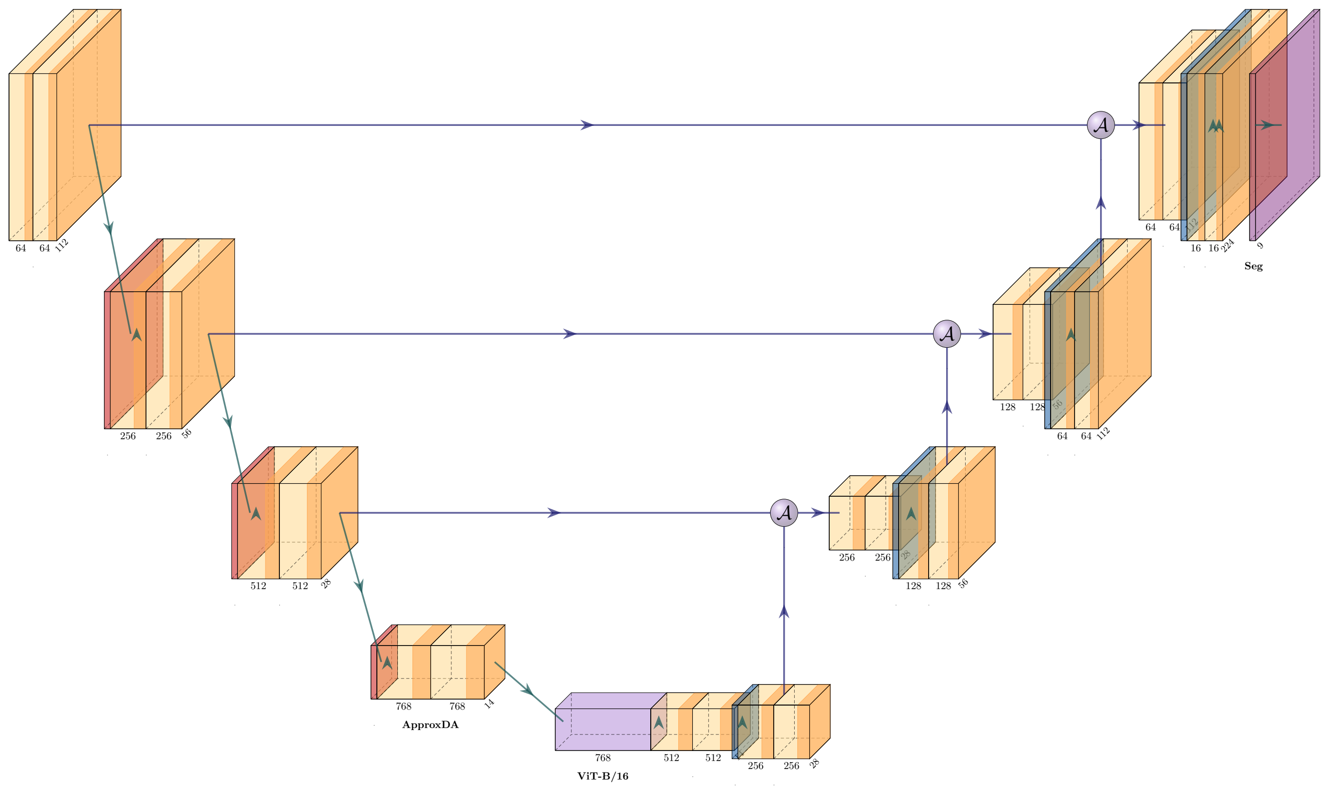


Fig. 1. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

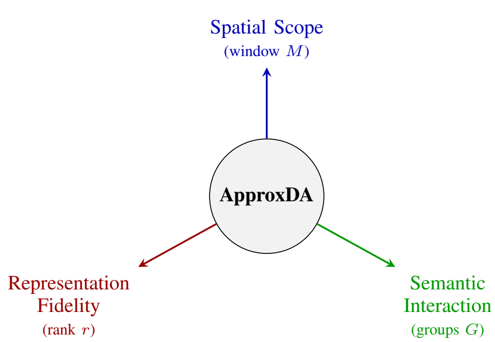


Fig. 2. The three orthogonal approximation axes of ApproxDA-TransUNet. Each axis is independently controllable, enabling clean ablations that isolate individual approximation effects.

trades long-range context for lower complexity. This axis most directly governs whether tasks with broader contextual requirements retain the anatomical context they depend on.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Semantic interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups reduces cross-channel dependency while retaining within-group semantic interaction.

These axes correspond to three distinct types of approximation: *spatial approximation* (how far attention reaches, controlled by M), *representation approximation* (how precisely affinity is computed, controlled by r), and *channel approximation* (how broadly channels interact, controlled by G). The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects. Figure 3 shows the resulting block structure, with both branches fused through the configurable gate.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA-TransUNet partitions the feature map into non-overlapping windows of size $M \times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [3], we partition the feature map into n_W non-overlapping $M \times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows $M{=}112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [4], query and key features inside each window are projected from $N_w{=}M^2$ to rank

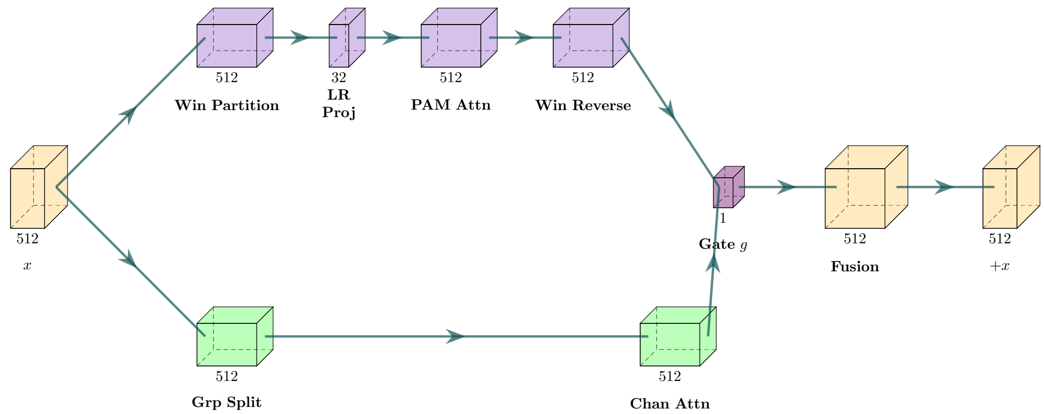


Fig. 3. ApproxDABlock internals. Input $\mathbf{x}$ splits into two branches: Low-Rank Windowed PAM (blue, upper) via window partition, low-rank projection, and attention; Grouped CAM (green, lower) via group split and channel attention. A learned gate g fuses the outputs; $\mathbf{x}$ is added as a residual.

$r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r \times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r \mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r \mathbf{D}^\top \in \mathbb{R}^{r \times C}. \tag{1}$$

The attention matrix $QK^\top \in \mathbb{R}^{N_w \times N_w}$ is approximated by $Q_r K_r^\top \in \mathbb{R}^{N_w \times r}$, where $Q_r, K_r \in \mathbb{R}^{N_w \times r}$. Total complexity is $\mathcal{O}(NrC)$. Combined, the two approximations reduce attention storage at 112×112 from $\approx$15 GB to $\approx$38 MB ($\sim$390$\times$).

These two hyperparameters—window size M and projection rank r—jointly constitute the spatial approximation operator and form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C\times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA-TransUNet approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \tag{2}$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [2] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f \Big(g \cdot \hat{\mathbf{P}} + (1-g) \cdot \hat{\mathbf{C}} \Big) + \mathbf{x}, \tag{3}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a 1×1 convolution, and g is determined by the `gate_mode` parameter:

`pam`:	$g = 1.0$	(PAM only)
`cam`:	$g = 0.0$	(CAM only)
`fixed`:	$g = 0.5$	(equal blend)
`learn`:	$g = \sigma(\mathbf{W}_g \cdot \mathrm{GAP}(\mathbf{x})) \in (0,1)^C$	

In gate=learn, $g \in (0,1)^C$ is a per-channel routing coefficient broadcast over all spatial positions; it collapses to a channel-wise blend rather than a spatially-adaptive or input-dependent scalar. This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA-TransUNet does not treat this routing module as a primary architectural contribution; rather, it serves as an analyzable component whose optimization dynamics—including an unexpected convergence to $g\approx0.5$ regardless of window size—are investigated in Section V.

F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}. \tag{4}$$

These approximations reduce layer-level attention complexity: windowed PAM from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NrC)$ ($r \ll N$); grouped CAM from $\mathcal{O}(C^2N)$ to $\mathcal{O}(C^2N/G)$. Since backbone operations dominate end-to-end computation, the primary value of ApproxDA-TransUNet is not whole-network FLOPs reduction but a controllable formulation for systematically isolating each approximation operator's effect on segmentation performance (Section IV).

IV. Results

We evaluate ApproxDA-TransUNet on three medical image segmentation benchmarks. Section V provides mechanistic analysis of the observed task-dependent patterns.

A. Experimental Setup

Datasets. We evaluate on three representative medical image segmentation benchmarks.

Synapse Multi-Organ CT [16]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes. Metrics: mean DSC and HD95.

Kvasir-SEG [17]: 1000 endoscopy polyp images (binary) with an 800/200 train/test split. Metrics: DSC and mIoU.

ISIC 2018 [18]: 2594 dermoscopy skin lesion images (binary) with a 2075/519 train/test split. Metrics: DSC and mIoU.

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K, trained on NVIDIA Tesla T4 (16 GB) GPUs. Optimizer: SGD, lr=0.01; batch size 24; input 224×224; 300 epochs; best checkpoint selected by evaluating every 15 epochs. Default ApproxDA-TransUNet configuration: $M{=}7$, $r{=}32$, $G{=}8$.

Gate Protocol. ApproxDA-TransUNet supports four gate modes: `pam` ($g{=}1.0$, PAM only), `cam` ($g{=}0.0$, CAM only), `fixed` ($g{=}0.5$, equal blend), and `learn` (g learned per channel). Unless otherwise specified, all main-result experiments use `gate=pam`. The gate ablation in Table VI evaluates all four modes to isolate the effect of routing strategy.

B. Comparison with State-of-the-Art

Tables I–III compare ApproxDA-TransUNet against published baselines across all three benchmarks.

Synapse. Table I benchmarks against 11 published methods. CNN-based models plateau at 76–78% DSC, with Att-UNet leading at 77.77%. Transformer-based methods progressively improve: TransUNet (77.48%), UCTransNet (78.23%), MIM (78.59%), Swin-Unet (79.13%), and DA-TransUNet (79.80%). ApproxDA-TransUNet with gate=pam, $M{=}28$, $r{=}32$ achieves **80.94%** DSC and 27.49 mm HD95—the highest DSC across all compared methods, surpassing DA-TransUNet by +1.14%. The intermediate window $M{=}28$ captures sufficient cross-organ context while limiting spurious long-range correlations; Section V analyzes this behavior in detail.

TABLE I
SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

Method	DSC (%) ↑	HD95 (mm) ↓
CNN-based		
U-Net [6]	76.85	39.70
U-Net++ [7]	76.91	36.93
Residual U-Net [9]	76.95	38.44
Att-Unet [8]	77.77	36.02
MultiResUNet [10]	77.42	36.84
Transformer-based		
TransUNet [11]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [19]	78.40	30.25
MIM [20]	78.59	26.59
Swin-Unet [12]	79.13	**21.55**
DA-TransUNet [2]	<u>79.80</u>	<u>23.48</u>
Dual-attention approximation (ours)		
ApproxDA-TransUNet (ours)	**80.94**	27.49

Baselines from [2] Table 1. ApproxDA: gate=pam, M=28, r=32, G=8.

Kvasir-SEG. On polyp segmentation (Table II), U-Net leads the CNN-based group (88.22% DSC) while TransUNet (87.91%) and DA-TransUNet† (88.44%) are the strongest Transformer-based entries. ApproxDA-TransUNet with gate=pam, $M{=}56$ achieves **90.17%** DSC, **84.27%** mIoU, and **44.35 mm** HD95—state-of-the-art across all three metrics, with a +1.73% DSC gain and −8.69 mm HD95 improvement over the nearest baseline.

TABLE II
SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2]†	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	**90.17**	**84.27**	**44.35**

Baselines from [2]; HD95 not originally reported. †Our DA-TransUNet rerun closely reproduces paper-reported values (88.47% DSC, 81.02% mIoU) and additionally provides HD95 (53.04 mm). ApproxDA: gate=pam, $M{=}56$, $r{=}32$.

ISIC 2018. On skin lesion segmentation (Table III), CNN-based methods cluster between 83–88% DSC, with TransUNet (88.78%) and DA-TransUNet (88.88%) leading among Transformer-based approaches. ApproxDA-TransUNet with gate=learn, $M{=}7$ achieves **89.58%** DSC (+0.70% over DA-TransUNet) and 82.66% mIoU, attaining the best DSC across all compared methods. The small optimal window and gate=learn configuration reflect that skin lesion segmentation is governed by local texture and boundary appearance rather than long-range spatial context, extending the low-GCS pattern observed on Kvasir-SEG to a third benchmark.

TABLE III
SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) ↑	mIoU (%) ↑
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	88.78	82.63
DA-TransUNet [2]	<u>88.88</u>	**82.78**
ApproxDA-TransUNet (ours)	**89.58**	<u>82.66</u>

All baselines from [2] Table 2. ApproxDA: gate=learn, $M{=}7$, $r{=}32$, $G{=}8$.

ApproxDA-TransUNet achieves state-of-the-art results across all three benchmarks (Figure 4). The markedly different optimal windows—$M{=}28$ for multi-organ CT vs. $M{=}7$ for skin lesion segmentation—combined with the $3.6\times$ window-sensitivity ratio between Synapse (2.30 pp DSC range) and Kvasir-SEG (0.64 pp), reveal a structured task-dependence that Section V examines in depth.

TABLE IV
EFFICIENCY COMPARISON ON SYNAPSE (NVIDIA TESLA T4)

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [2]	107.95	30.2	11.5 G	11.4h	120.0	No†
ApproxDA-TransUNet (gate=pam, M=7)	112.98	32.1	6.4 G (×2 GPU)	8.3h (×2 GPU)	121.3	Yes (DDP)
ApproxDA-TransUNet (gate=learn, M=7)	114.90	32.1	10.6 G	11.5h	114.2	Yes (DDP)

†DA-TransUNet does not support multi-GPU training. ApproxDA-TransUNet uses Distributed Data Parallel (DDP); per-GPU batch 12, LR unchanged.

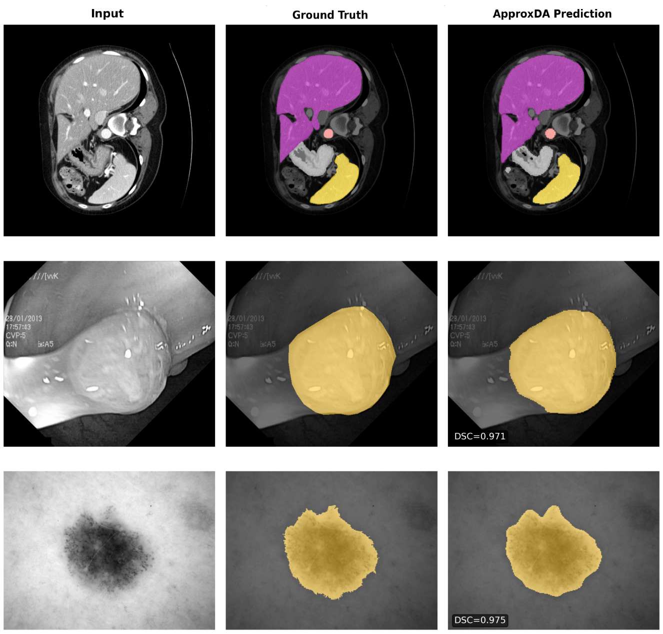


Fig. 4. Qualitative segmentation comparison. **Top (Synapse, M=28):** multi-organ CT; ApproxDA 80.94% DSC (+1.14%). **Middle (Kvasir-SEG, M=56):** polyp; 90.17% DSC (+1.73%). **Bottom (ISIC 2018, M=7):** skin lesion; 89.58% DSC (+0.70%).

C. *Efficiency Analysis*

Table IV shows that the ViT backbone accounts for the majority of compute overhead, leaving attention approximation with little room for end-to-end hardware savings (GFLOPs: 32.1 vs. 30.2); the main practical benefit is DDP compatibility—DA-TransUNet fails with a runtime error under multi-GPU execution, while ApproxDA-TransUNet runs natively.

V. ABLATION STUDIES AND MECHANISTIC ANALYSIS

This section investigates why ApproxDA-TransUNet behaves differently under different approximation settings and segmentation tasks through a series of controlled ablation studies. Beyond validating the proposed design, these experiments provide mechanistic insights into attention approximation and its task-dependent behavior.

A. *Approximation Strength Ablation*

We first investigate how approximation strength influences segmentation performance by varying the spatial window size and approximation rank while keeping all other components unchanged. Table V isolates this effect by comparing $M \in \{7, 14, 28, 56, 112\}$ under gate=pam, r=32 on Synapse. Approximation is present in all rows—the right approximation *scale*, not the absence of approximation, governs performance.

TABLE V
SPATIAL APPROXIMATION ABLATION ON SYNAPSE (G=8)

Config	M	r	Gate	DSC (%) ↑	HD95 (mm) ↓
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn†	78.04	29.09
Windowed + low-rank	28	32	pam	**80.94**	<u>27.49</u>
Windowed + low-rank	56	32	pam	78.90	35.23
Global + low-rank	112	32	pam	<u>79.44</u>	**27.26**

†Gate collapsed to $g \approx 0.5$. M=112 via window clamping yields full-map attention at all scales. Non-monotonic peak at M=28: both under-context (M=7) and over-context (M=112) underperform.

The Synapse DSC-vs-M curve is *non-monotonic*: peak at M=28 (80.94%, +1.14% vs. DA-TransUNet), with all other scales underperforming—too-local (M=7: 78.64%), intermediate-upper (M=56: 78.90%), and fully-global (M=112: 79.44%). This pattern is analogous to the classical bias-variance tradeoff: under-context windows have high bias (missing necessary dependencies); over-context windows introduce spurious variance (irrelevant distant tokens); the intermediate M=28 achieves the optimal balance. The DSC range of 2.30 pp across M values confirms high window sensitivity on this task.

The M=14/r=64 row shows that increasing rank does not compensate for gate collapse ($g \approx 0.5$, Section V-C): DSC 78.04% is lower than the gate=pam M=7/r=32 baseline (78.64%), suggesting r=32 already captures sufficient attention information and rank is not the performance bottleneck.

Preserving sufficient spatial receptive field is more important than increasing approximation rank; the optimal window scale—not approximation rank—governs segmentation performance.

B. *Cross-task Ablation and Global Context Sensitivity*

We next examine whether the same approximation strategy behaves consistently across different segmentation tasks.

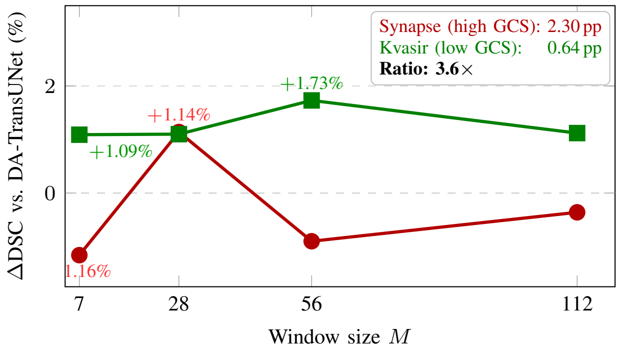


Fig. 5. Per-window gain of ApproxDA-TransUNet over DA-TransUNet (ΔDSC, gate=pam, r=32). The Synapse curve (red) spans 2.30 pp and crosses zero: wrong window choices actively hurt on this empirically high-GCS task. The Kvasir-SEG curve (green) spans only 0.64 pp and stays positive throughout: any reasonable window helps on this empirically low-GCS task. The 3.6× span ratio (annotation box) directly operationalizes empirical GCS.

Although ApproxDA-TransUNet uses identical approximation operators across all experiments, its sensitivity to window selection varies substantially. On Synapse multi-organ CT, the DSC-vs-M curve (Figure 5) is non-monotonic with a sharp peak at $M{=}28$ and a 2.30 pp range across the measured window sizes. On Kvasir-SEG, the curve is nearly flat (0.64 pp range) with a shallow peak at $M{=}56$, indicating the task is largely window-robust. On ISIC 2018, ApproxDA-TransUNet (gate=learn, $M{=}7$) achieves 89.58% DSC vs. 88.88% for DA-TransUNet (+0.70%), consistent with the low-sensitivity pattern. With task-appropriate windows, gains are positive across all three benchmarks: +1.14% (Synapse, $M{=}28$), +1.73% (Kvasir, $M{=}56$), and +0.70% (ISIC, $M{=}7$).

These observations suggest that segmentation tasks exhibit different degrees of **Global Context Sensitivity (GCS)**—the degree to which accurate segmentation depends on preserving large-scale spatial context. Since GCS is not directly observable, we estimate it empirically through the performance response to window-size ablations, summarized by the DSC range:

$$\Delta\text{DSC} = \max_M \text{DSC}(M) - \min_M \text{DSC}(M),$$

measured under fixed gate=pam, r=32. For Synapse, ΔDSC = 2.30 pp; for Kvasir-SEG, 0.64 pp—a 3.6× ratio confirming that Synapse exhibits substantially higher empirical GCS. Tasks with higher empirical GCS exhibit steep DSC-vs-M curves with a well-defined optimal scale; lower-GCS tasks show broad performance plateaus where any reasonable window achieves competitive performance. Crucially, high GCS does not imply approximation is harmful: with the right window, ApproxDA-TransUNet outperforms full dual attention on *both* tasks. GCS characterizes how *sensitive* performance is to window choice—not the direction of the effect.

The task-level differences in GCS are grounded in the nature of each segmentation problem. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. The high sensitivity to window size reflects that both too-local windows (insufficient cross-organ context) and fully-global windows (noise from unrelated background) degrade performance; the intermediate scope ($M{=}28$) provides the optimal balance. In contrast, polyp and skin lesion segmentation are primarily governed by local appearance, boundary contrast, and texture consistency. For these low-GCS tasks, attention approximation introduces an *inductive bias* favoring local interactions while suppressing unnecessary long-range dependencies—a prior that better matches the intrinsic structure of the problem.

We validate this locality bias hypothesis via controlled attention-map analysis: comparing gate=pam $M{=}7$ (windowed) versus $M{=}112$ (global) attention effects on 10 Kvasir polyp images, windowed attention concentrates 62.3% of its effect on the polyp region versus only 34.2% for global attention—a 1.82× difference confirmed in 9/10 samples (Figure 6). Windowed attention focuses on polyp boundaries while global attention disperses to surrounding tissue, providing direct evidence for the inductive bias alignment hypothesis.

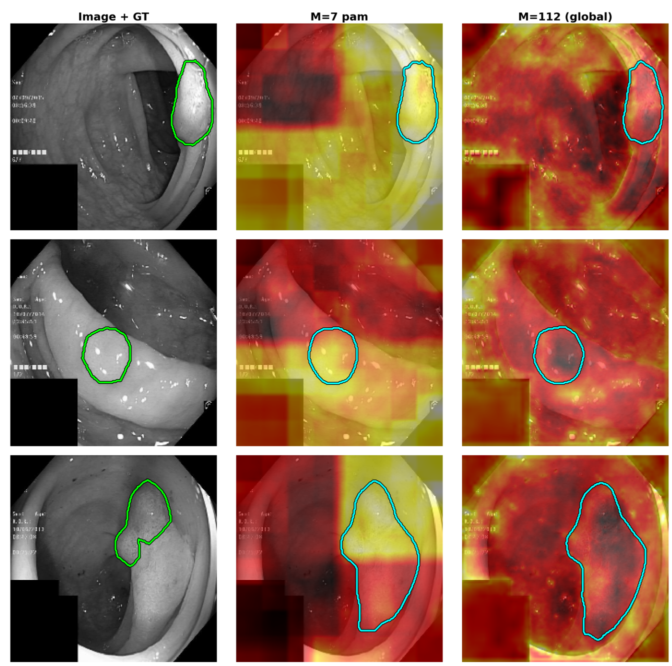


Fig. 6. Attention-map analysis on three representative Kvasir-SEG polyp images (selected from 10; samples with largest M=7 vs. M=112 on-mask gap). Each row: input image with GT contour (left), windowed PAM heatmap ($M{=}7$, gate=pam; center), global PAM heatmap ($M{=}112$, gate=pam; right). Windowed attention concentrates 62.3% of its effect on the polyp region vs. 34.2% for global attention—a 1.82× difference confirmed in 9/10 samples. Heatmap intensities reflect $\|\text{output} - \text{input}\|_2$ averaged across decoder scales and upsampled to 224×224.

Together, these findings reframe the role of approximation: not as a capability reduction but as a task-dependent inductive bias adjustment whose value depends on how well the enforced locality prior aligns with the task's intrinsic context scale.

Attention approximation should be selected according to the contextual characteristics of the target task rather than applied uniformly: high-GCS tasks require careful window tuning;

low-GCS tasks are largely window-robust and benefit from the locality prior that windowed approximation enforces.

C. Gate Ablation and Mechanistic Analysis

Finally, we investigate whether adaptive routing contributes to approximation performance by comparing PAM-only, CAM-only, and learnable gate configurations. Table VI reports results for all four gate modes on Synapse (M=7, r=32).

TABLE VI
GATE MODE ABLATION ON SYNAPSE (M=7, r=32 UNLESS NOTED)

Gate Mode	g	DSC (%) ↑	HD95 (mm) ↓
gate=pam	1.0	**78.64**	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn (M=14, r=64)	≈0.5*	78.04	**29.09**
gate=learn	≈0.5*	77.78	34.29

All rows: ApproxDA-TransUNet. *Gate collapsed to g≈0.5 throughout training. **Bold**: best; <u>underline</u>: second best per column.

Gate=pam (78.64%) outperforms gate=cam (78.26%) and both gate=learn configurations (77.78% at M=7; 78.04% at M=14, r=64). Notably, gate=learn with a larger window and higher rank (M=14, r=64) still underperforms gate=pam at M=7 (78.04% vs. 78.64%), confirming that gate collapse is more damaging than a suboptimal window choice. In both gate=learn configurations, the gate converges to g≈0.5 regardless of window size, producing a fixed 50/50 blend that performs below both single-branch modes.

Figure 7 provides direct evidence: (a) the gate distribution across all validation slices is nearly constant at g≈0.5 regardless of input, and (b) gate values show no correlation with feature entropy, ruling out sample-adaptive routing. This collapse is identical at both M=7 and M=14, confirming it is a fundamental property of symmetric two-branch gating rather than a windowing artifact.

Optimization intuition. Given fused output $\mathbf{F} = g\mathbf{F}_{\mathrm{P}} + (1-g)\mathbf{F}_{\mathrm{C}}$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}}) \,. \quad (5)$$

At initialization g≈0.5, both branches receive equal gradient $\frac{1}{2}\partial\mathcal{L}/\partial\mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_{\mathrm{P}} \approx \mathbf{F}_{\mathrm{C}}$, so Eq. (5)≈ $\mathbf{0}$. This is self-reinforcing: equal gradients → convergent representations → near-zero gate gradient → g fixed at 0.5—a stable equilibrium of symmetric dual-branch optimization.

Empirical evidence. After 300 epochs, the gate range at M=7 is $[0.4993, 0.5010]$ with Δg=0.0017. At M=14: g≈0.5002 with Δg≈0.0000.

Simple scalar gating cannot provide effective adaptive routing because branch symmetry prevents meaningful specialization. Future adaptive attention mechanisms should explicitly encourage branch diversity—through asymmetric initialization, diversity-encouraging loss terms, or non-symmetric routing—rather than relying solely on learnable fusion weights.

Summary of Findings

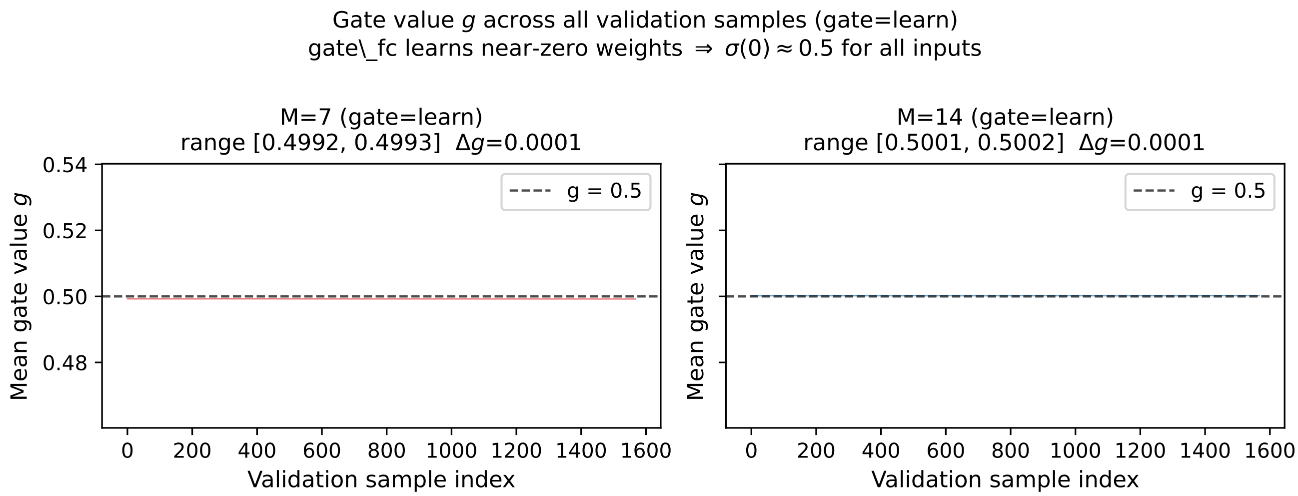
Overall, the ablation studies reveal three important findings. First, approximation strength should primarily preserve spatial receptive field rather than aggressively compress feature representations: intermediate windows outperform both too-local and too-global configurations, and increasing rank yields no additional benefit. Second, approximation effectiveness depends on task-specific contextual requirements (GCS): high-GCS tasks such as multi-organ CT require careful window tuning while low-GCS tasks such as polyp and skin lesion segmentation are window-robust, with a 3.6× sensitivity ratio confirming this distinction. Third, current scalar gating collapses to a stable degenerate equilibrium due to gradient symmetry, contributing little to adaptive routing. Together, these findings explain why ApproxDA-TransUNet consistently improves performance and provide practical guidelines for future attention approximation design.

D. Limitations

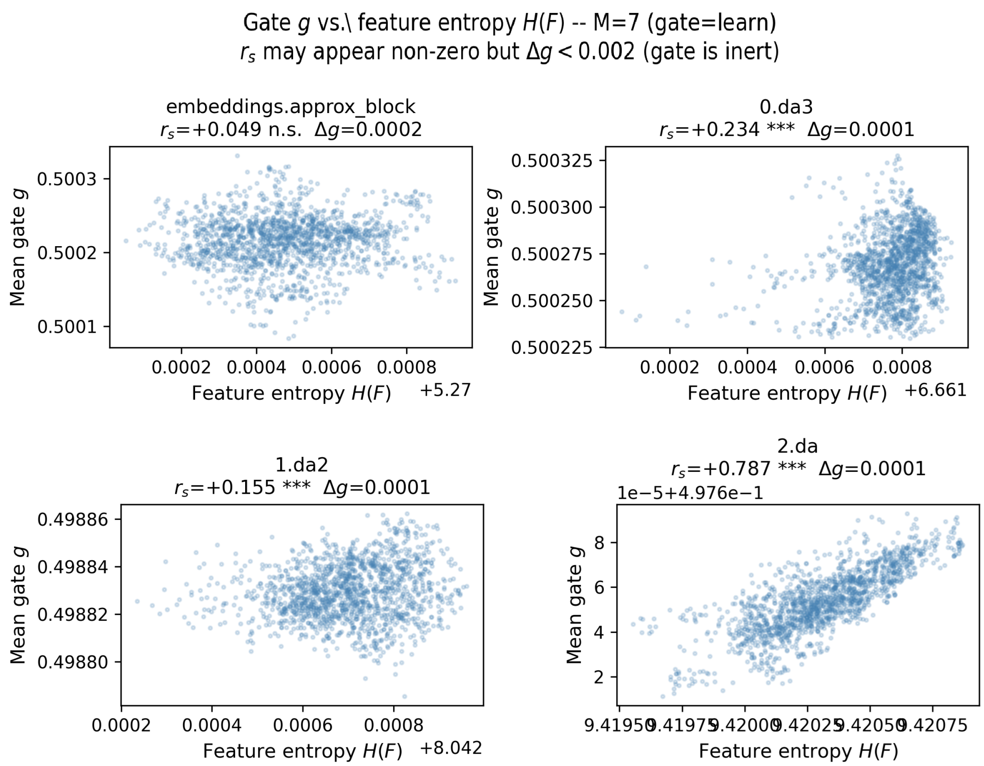
Although the ΔDSC measure provides a computable GCS estimate from the window sensitivity ablation, a rigorous GCS estimator that generalizes across imaging modalities and training protocols—without requiring the ablation itself—remains to be developed. A natural candidate proxy, center-cropping the input to restrict visible context, is invalidated by a more fundamental issue: evaluation against the full-resolution label penalises the crop for missing structures that lie outside the crop window, regardless of the model's actual context reasoning. For randomly-positioned targets (polyps, lesions), small crops frequently exclude the target entirely; for anatomically-fixed but spatially-extended targets (abdominal organs), crops exclude large organ fractions. In both cases DSC collapses not because the model lacks global reasoning but because the crop removes the target itself. Window sensitivity within the model—where target visibility is unchanged—is the appropriate substitute. Additionally, the cross-task comparison covers three benchmarks (Synapse, Kvasir-SEG, and ISIC 2018); the GCS hypothesis should be validated on a broader range of medical segmentation tasks before being applied prescriptively. Finally, the two mechanisms—inductive bias alignment and context interference—are empirically motivated but not formally proven; their relative contributions and the conditions under which each dominates warrant further investigation.

VI. CONCLUSION

This paper presented **ApproxDA-TransUNet**, an efficient dual-attention approximation framework that replaces computationally expensive global dual attention with controllable spatial and channel approximations. Extensive experiments on three representative medical image segmentation benchmarks demonstrate that ApproxDA consistently outperforms the original DA-TransUNet while providing substantially improved computational efficiency. These results indicate that



(a) Gate distribution across validation slices



(b) Entropy–gate correlation

Fig. 7. Gate collapse analysis (Synapse, gate=learn). **(a)** Gate value g across all validation slices for $M{=}7$ and $M{=}14$. Both produce near-constant $g{\approx}0.5$ (range: $[0.4993,\ 0.5010]$ at $M{=}7$; $g{\approx}0.5002$ at $M{=}14$), confirming the gate is inert regardless of input or window size. **(b)** Gate value vs. per-block feature entropy $H(F)$ ($M{=}7$) for all four ApproxDABlock locations: `embeddings.approx_block` (768 ch, 14×14, pre-ViT patch block), `blocks[0].da3` (512 ch, 28×28, deepest decoder skip), `blocks[1].da2` (256 ch, 56×56, middle decoder skip), and `blocks[2].da` (64 ch, 112×112, shallowest decoder skip). All four blocks cluster at $g{\approx}0.5$ with Spearman $r_s{\approx}0$, confirming gate collapse is uniform across scales and architectural positions.

carefully designed attention approximation can preserve, and even enhance, segmentation performance without relying on full global attention.

Beyond the proposed architecture, this work provides a deeper understanding of **when** and **why** attention approximation is effective. Through comprehensive ablation studies, we show that preserving sufficient spatial receptive field is considerably more important than increasing approximation rank. Cross-task analyses further reveal that the effectiveness of attention approximation depends on the contextual characteristics of the target task. We describe this empirical phenomenon as **Global Context Sensitivity (GCS)**, where tasks requiring stronger global reasoning exhibit greater sensitivity to the approximation scale, whereas tasks dominated by local appearance remain comparatively robust.

Our gate analysis further demonstrates that the commonly adopted learnable scalar routing mechanism consistently collapses toward nearly uniform fusion because of gradient symmetry between the spatial and channel attention branches. This observation suggests that simply introducing adaptive fusion weights is insufficient to achieve effective dynamic routing, motivating future attention mechanisms that explicitly encourage branch specialization.

Overall, our findings suggest that attention approximation should not be regarded merely as an efficiency technique, but as a **task-dependent inductive bias** whose effectiveness is fundamentally governed by the contextual requirements of the segmentation task. We hope this work encourages future research to move beyond designing increasingly efficient approximation modules toward developing a more principled understanding of attention approximation across different vision tasks.

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